

CyberOps Associate - Skills Assessment version 2026

Introduction

You have been hired as a junior security analyst. As part of your training, you were tasked to determine any malicious activity.

The associated network segment:

- **LAN: 10.22.15.0/24**
- Default gateway: 10.22.15.1
- Domain controller: 10.22.15.2

Incident happened between 6th and 8th November 2018.

You will have access to the internet to learn more about the events. You can use websites, such as VirusTotal or PacketTotal, to upload and verify threat existence.

The tasks below are designed to provide some guidance through the analysis process.

You will practice and be assessed on the following skills:

- o Evaluate event alerts using Squil and Kibana.
- o Use Wireshark to examine details and export object representing malware.
- o Use VirusTotal or PacketTotal to upload and verify threat existence.

Required Resources

- **either** custom Security Onion virtual machine
- **or** PCAP file imported to Security Onion using **so-import-pcap** command
- Internet access

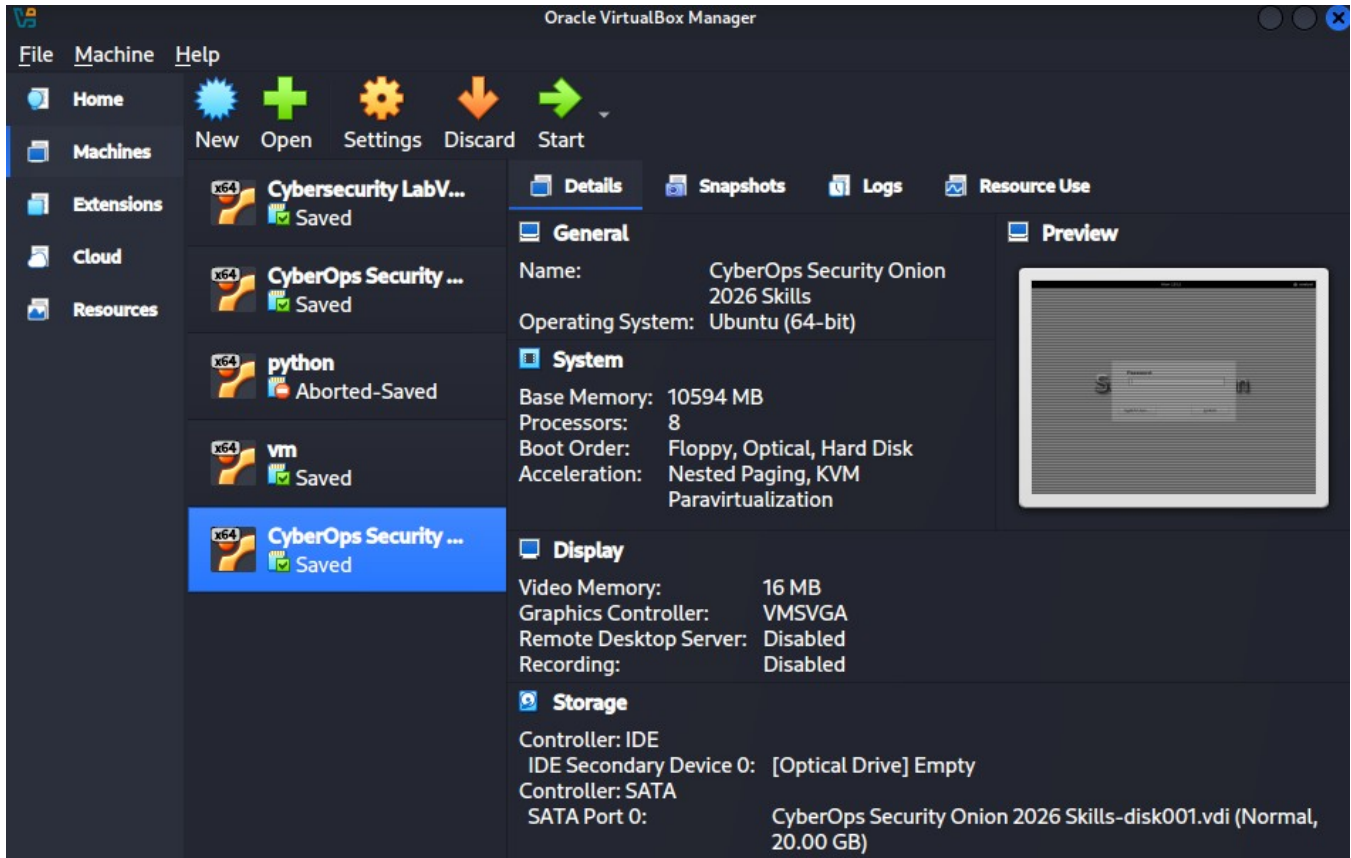
Instructions

Part 1: Gather the Basic Information

In this part, you will review the alerts listed in Security Onion VM and gather basic information for the interested time frame.

Step 1: Verify the status of services

- a. Log into Security Onion VM using with the username **analyst** and password **cyberops**.
- b. Open a terminal window. Enter the **sudo so-status** command to verify that all the services are ready.
- c. When the nsm service is ready, log into Sguil or Kibana with the username **analyst** and password **cyberops**.



We downloaded the specific VM machine and connected to it via our Oracle VirtualBox Manager.

Step 2: Gather basic information.

- a. List the alerts noted during this time frame associated with the malware. Optionally you can update Suricata rules with Emerging Threats PRO/OPEN Ruleset ruleset and see more detailed alerts in Sguil.

The malware triggered these alerts:

- ET POLICY DNS Update from external net
- ET INFO EXE – Served Attached HTTP
- ET POLICY PE EXE or DLL Windows file download HTTP
- ET INFO Executable Retrieved with Minimal HTTP headers – potential second stage downloaded
- ET POLICY External IP lookup domain

The screenshot shows the Sguil-0.9.0 interface running in a virtual machine. The main window displays a list of alerts under the 'RealTime Events' tab. The alerts are sorted by date and time, with the most recent at the top. The interface includes a menu bar with 'File', 'Query', 'Reports', 'Sound', 'Off', 'ServerName: localhost', 'UserName: analyst', 'UserID: 2', and a timestamp '2026-05-04 13:14:33 GMT'. Below the menu bar, there are tabs for 'RealTime Events' and 'Escalated Events'. The alert list has columns for ST, CNT, Sensor, Alert ID, Date/Time, Src IP, SPort, Dst IP, DPort, Pr, and Event Message. The bottom section of the interface contains a 'System Msg' tab with fields for IP Resolution, Agent Status, Snort Statistics, and System Msg. It also includes a 'Show Packet Data' section with a table for packet details, including Source IP, Dest IP, Ver, HL, TOS, len, ID, Flags, Offset, TTL, ChkSum, and a 'Search Packet Payload' section with radio buttons for Hex, Text, and NoCase.

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	17	seconion-...	5.234	2019-07-19 18:53:12	172.16.4.205	49249	185.243.115.84	80	6	ET POLICY Data POST to a...
RT	114	seconion-...	5.251	2019-07-19 18:57:23	172.16.4.205	49255	31.7.62.214	443	6	ET POLICY HTTP traffic on ...
RT	2	seconion-...	5.365	2020-02-21 00:53:55	172.17.8.174	62362	172.17.8.8	53	17	ET POLICY DNS Update Fro...
RT	13	seconion-...	5.366	2020-02-21 00:55:07	49.51.172.56	80	172.17.8.174	49731	6	ET CURRENT_EVENTS Lik...
RT	13	seconion-...	5.379	2020-02-21 00:55:07	49.51.172.56	80	172.17.8.174	49731	6	ET CURRENT_EVENTS Win...
RT	13	seconion-...	5.392	2020-02-21 00:55:07	49.51.172.56	80	172.17.8.174	49731	6	ET POLICY PE EXE or DLL ...
RT	4	seconion-...	5.406	2020-02-21 01:11:48	91.211.88.122	443	172.17.8.174	49760	6	ET TROJAN ABUSE.CH SS...
RT	1	seconion-...	5.1	2020-06-11 03:41:20	209.165.200.235	6200	209.165.201.17	45415	6	GPL ATTACK_RESPONSE i...
RT	351	seconion-...	1.1	2020-06-19 18:09:28	0.0.0.0		0.0.0.0	0	0	[OSSEC] File added to the s...
RT	23	seconion-...	1.2	2020-06-19 18:09:29	0.0.0.0		0.0.0.0	0	0	[OSSEC] Integrity checksum...
RT	7	seconion-...	1.4	2020-06-19 18:10:04	0.0.0.0		0.0.0.0	0	0	[OSSEC] New group added t...
RT	7	seconion-...	1.5	2020-06-19 18:10:04	0.0.0.0		0.0.0.0	0	0	[OSSEC] New user added to ...
RT	2	seconion-...	1.18	2020-06-19 18:14:41	0.0.0.0		0.0.0.0	0	0	[OSSEC] Listened ports stat...

- b. Determine the exact date and time when the malicious traffic began.

So we will check the first alert that has occurred in our Sguil interface (7th .11.2018 20:40:54)

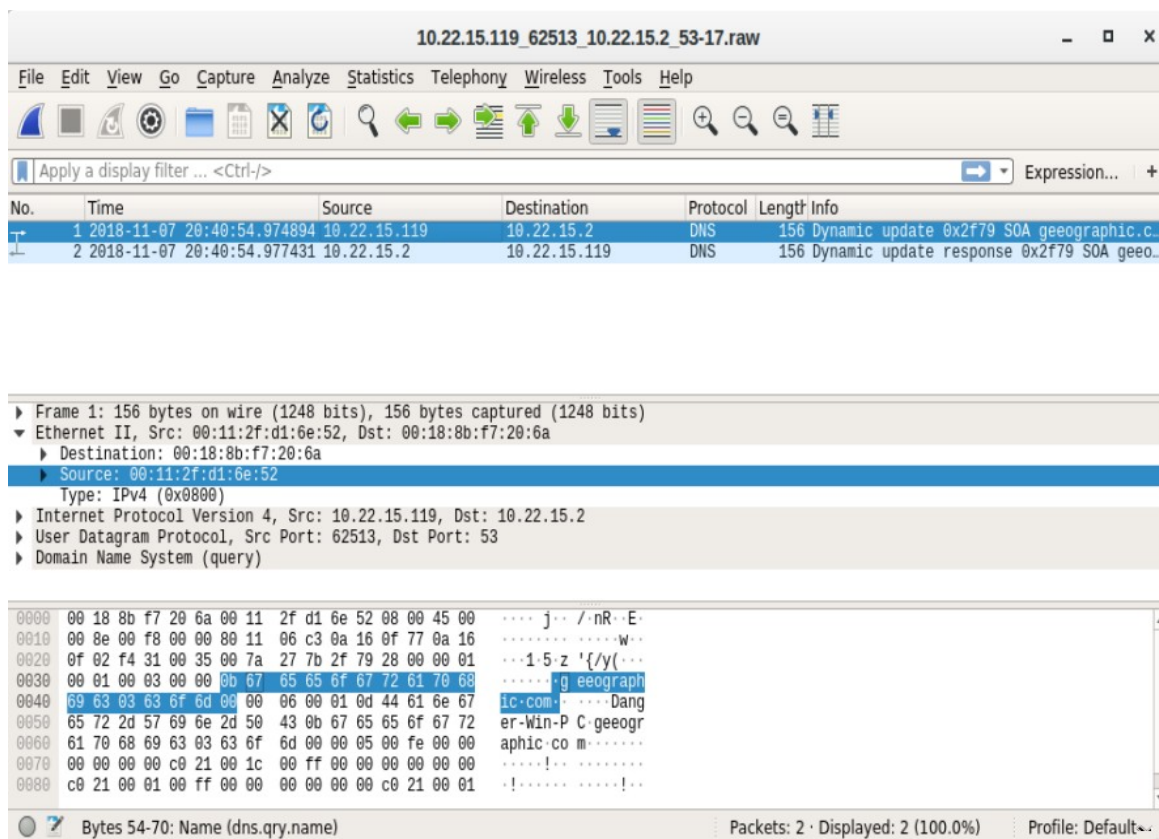
ST	CNT	Sensor	Alert ID	Date/Time	Δ	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	5	seconion-...	5.150	2018-08-11 05:20:59		149.129.222.112	80	192.168.1.95	49335	6	ET INFO Packed Executable...
RT	5	seconion-...	5.155	2018-08-11 05:20:59		149.129.222.112	80	192.168.1.95	49335	6	ET POLICY PE EXE or DLL ...
RT	5	seconion-...	5.160	2018-08-11 05:20:59		149.129.222.112	80	192.168.1.95	49335	6	ET INFO Executable Retrie...
RT	66	seconion-...	5.165	2018-08-11 05:30:06		192.168.1.95	50185	185.68.93.18	80	6	ET INFO GENERIC SUSPIC...
RT	1	seconion-...	5.1236	2018-11-07 20:40:54		10.22.15.119	62513	10.22.15.2	53	17	ET POLICY DNS Update Fro...
RT	12	seconion-...	5.1237	2018-11-07 20:47:11		46.29.160.132	80	10.22.15.119	49208	6	ET INFO EXE - Served Atta...
RT	12	seconion-...	5.1249	2018-11-07 20:47:11		46.29.160.132	80	10.22.15.119	49208	6	ET POLICY PE EXE or DLL ...
RT	12	seconion-...	5.1261	2018-11-07 20:47:11		46.29.160.132	80	10.22.15.119	49208	6	ET INFO Executable Retrie...
RT	2	seconion-...	5.1273	2018-11-07 20:49:57		10.22.15.119	64497	208.67.222.222	53	17	ET POLICY External IP Look...
RT	1	seconion-...	5.439	2019-03-19 01:45:03		10.0.90.215	52609	10.0.90.9	53	17	ET POLICY DNS Update Fro...
RT	2	seconion-...	5.440	2019-03-19 01:47:04		10.0.90.215	49204	209.141.34.8	80	6	ET INFO Executable Downlo...
RT	1	seconion-...	5.441	2019-03-19 01:47:04		10.0.90.215	49204	209.141.34.8	80	6	ET CURRENT_EVENTS Pos...
RT	2	seconion-...	5.442	2019-03-19 01:47:04		209.141.34.8	80	10.0.90.215	49204	6	ET CURRENT_EVENTS Lik...

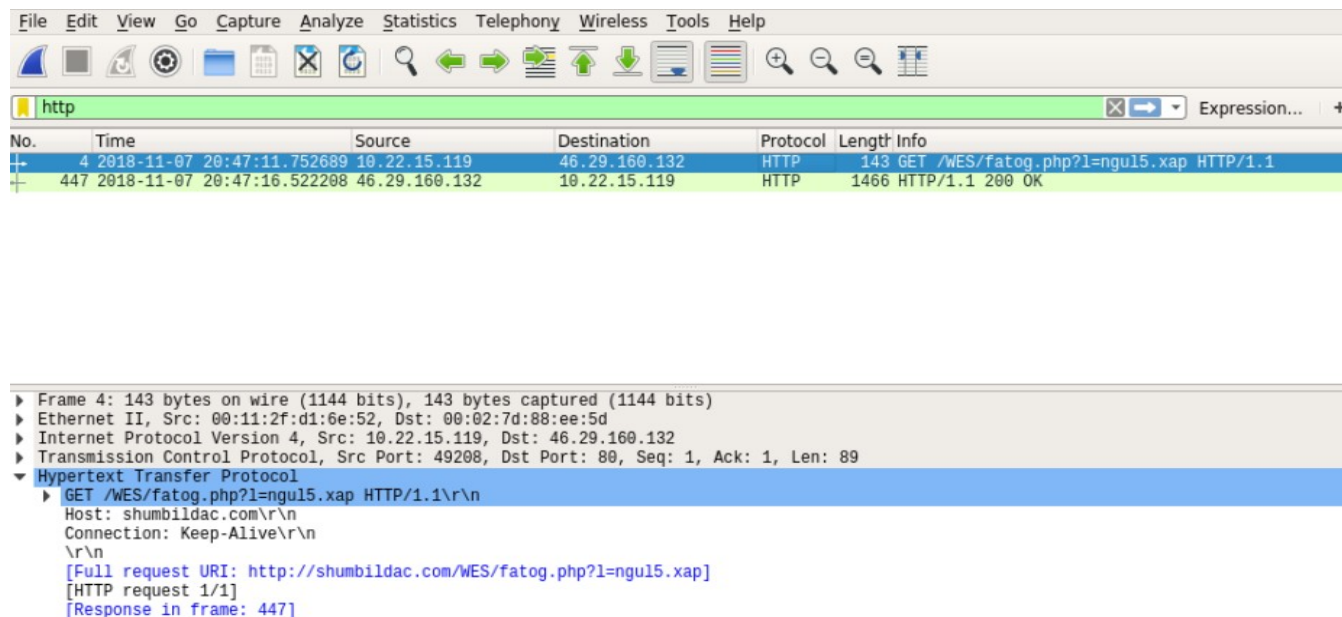
- c. Write IP address of host worthy of investigation (hint: 10.22.15.*** where *** are digits of last byte).

The host 10.22.15.119 is worth of investigation because he/she started the malicious activity or more precisely the malicious activity started on their workstation.

- d. Identify the MAC address of the compromised Windows host.

We will identify the MAC address of the compromised Windows host in Wireshark. Which is 00:11:2f:d1:6e:52.





- h. Try to extract malware from PCAP file, then compute its SHA256 hash and pass it to VirusTotal. Write down both hash and your findings.

To extract object we click on File > Export Objects > HTTP and we can see the filename in the list of files so we choose save and then in Terminal we use command sha256sum to convert it into hash file.

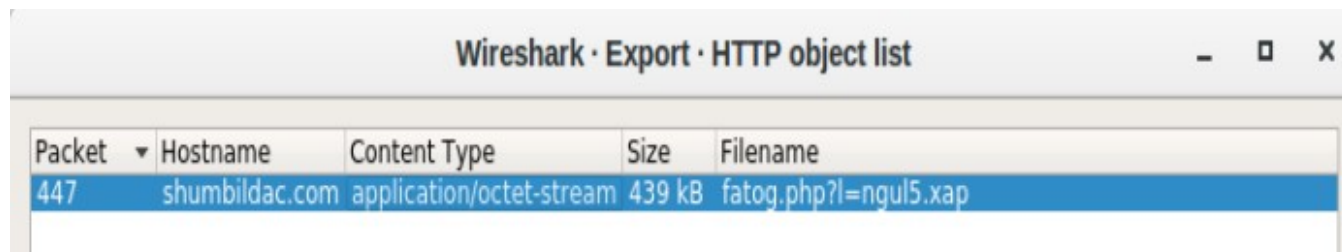
Hash: 97f149f146b0ec63c32abff204ae27638f0310536172b0f718f1a91a5672fe71

VirusTotal findings: 59/72 security vendors flagged this file as malicious

Original name: scaleopen.exe

Threat categories: trojan , banker

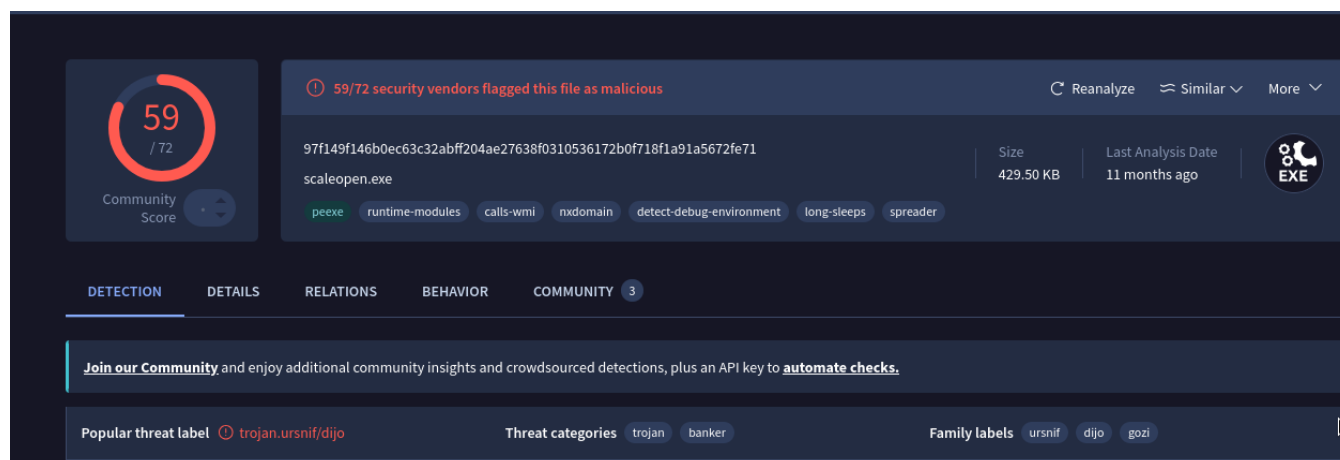
Family labels: ursnif, dijo, gozi



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analyst@SecOnion: ~/Desktop
File Edit View Search Terminal Help
analyst@SecOnion:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
analyst@SecOnion:~$ cd Desktop/
analyst@SecOnion:~/Desktop$ ls
fatog.php%3f1=ngul5.xap          securityonion-readme.desktop
securityonion-cyberchef.desktop  securityonion-sguil.desktop
securityonion-kibana.desktop     securityonion-squert.desktop
analyst@SecOnion:~/Desktop$ sha256sum fatog.php%3f1=ngul5.xap
97f149f146b0ec63c32abff204ae27638f0310536172b0f718f1a91a5672fe71  fatog.php%3f1=ngul5.xap
analyst@SecOnion:~/Desktop$

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i) Classify the type of malware contained in the downloaded Windows executable. Google for any details.

The type of the malware is trojan and goes under the name of Ursnif, which is one of the most common active banking trojans there is, which steals login credentials, personal data and information about the bank accounts – it is spread by phishing emails and compromised attachments.

j) List as much indicators of compromise as possible (e.g., IP addresses, ports, and domains associated with the malware).

Infected machine's IP address – 10.22.15.119

IP address of attacker's C2 server – 46.29.160.132:80

Malicious domain – <http://shumbildac.com/WES/fatog.php?1=ngu15.xap>, where attacker downloaded the banking trojan and the trojan then sent the stolen credentials via the DNS to 208.67.222.222.

Connection with attacker's domain on (DNS Exfiltration)– 208.67.222.222:53

Report your findings.

Summarizes your findings based on the information you have gathered from the previous parts, summarize your findings. Write three sentences: 1) when it happened and to whom; what is the source; 2) propose counter measures and next steps.

1) The alerts started popping up on 7th of November 2018 at 20:40:54 on the workstation of an employee - Carlos Danger, when the workstation updated DNS server (10.22.15.2), which is common work of either:

- Persistence
- C2 discovery
- DNS tunneling
- DNS hijacking

Then after the update which enabled control for an attacker (46.29.160.132), he started downloading the banker malware from **<http://shumbildac.com/WES/fatog.php?1=ngu15.xap>**, which he had done successfully and in the last step of the alert we can see that the workstation sent the credentials encrypted as a DNS to IP address 208.67.222.222.

KEY POINT: We still don't know when and how the attack started (don't have the alerts on it), but the potential intrusion could be by phishing email or worm, but in case of worm the malware would spread into all devices, so the biggest possibility is Carlos Dangers negligence in opening on malicious attachments and succumbing to social engineering attack.

2) Countermeasures should go accordingly:

- At first isolate the Carlos Danger's machine from the rest of the network so the attacker will lose connection to it + it won't spread elsewhere.
- Don't turn off the computer so you can still work with the RAM image for forensics analysis investigation.
- Block the IP addresses connected to this activity (46.29.160.132, 208.67.222.222) as well as the malicious domain, where the attacker downloaded the malware file.
- Check the rest of the network/devices for potential intrusions and notify the employees of the situation.
- Create disciplinary session with Carlos danger and investigate the previous actions to better correlate and understand the attack chain. > Update lessons learnt